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Concrete Experience





United States Army Cyber Center of Excellence

Electromagnetic Warfare Platoon (PLT) Force Structure and Mission

112-17ABOLC

OVERALL CLASSIFICATION:

UNCLASSIFIED



Learning Step Activities



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- LSA 1 - Describe the EW PLT force structure
- LSA 2 - Define EW PLT Mission
- LSA 3 - Compare EW PLT Missions by organization
- LSA 4 - Identify Key EW PLT Doctrine



Administrative Information



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Safety Requirements NONE

Risk Assessment Level LOW

Environmental Considerations None

Evaluation None



References



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- FM 3-12. *Cyberspace and Electromagnetic Warfare Operations*. 17 Sep 2025.
- ATP 3-12.3, C1. *Electromagnetic Warfare Techniques*. 18 Sep 2024.
- ATP 3-12.4. *Electromagnetic Warfare Platoon*. 09 Jan 2023.



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Describe the EW PLT force structure



EW PLT Overview

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- The EW platoon is a 14-40* personnel element composed of MOS 17 series soldiers with mounted or dismounted equipment that support their unit's ability to:
 - Target with lethal and nonlethal effects
 - Improve situational awareness and evaluation of the EMS
 - Conduct Intelligence surveillance and reconnaissance



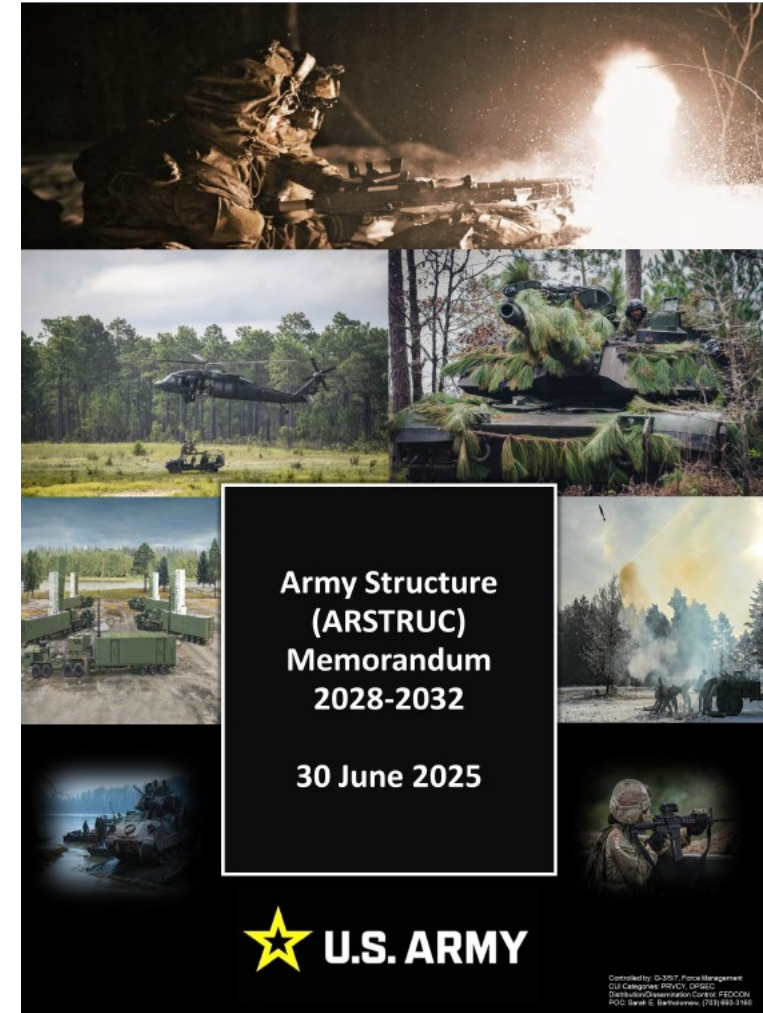


PE- ARSTRUC



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- Task
 - Research ARSTRUC and brief significant changes for 17 series
 - 3 Groups
 - EW
 - Cyber
 - MDTF

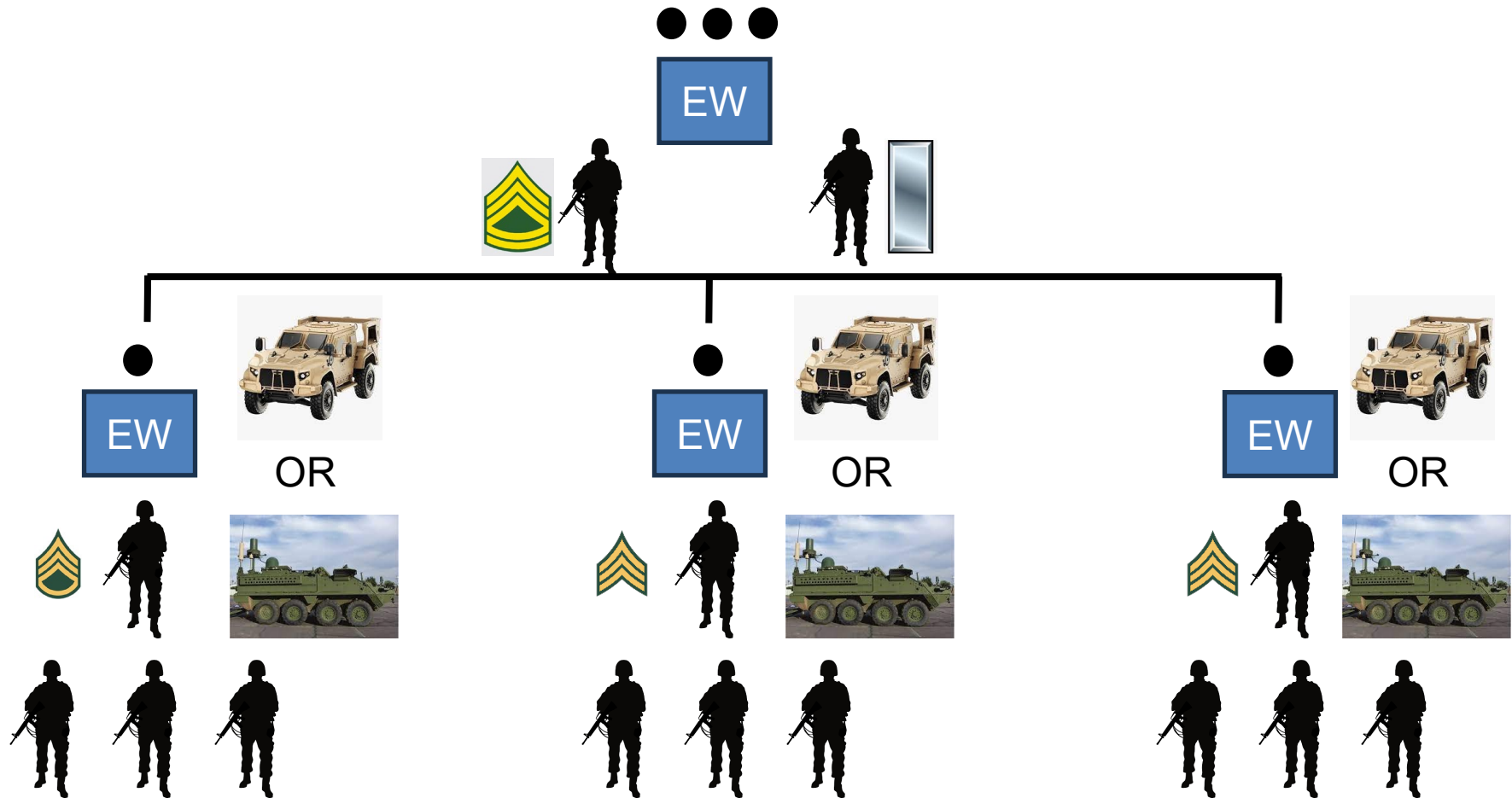




ABCT/SBCT EW PLT Force Structure



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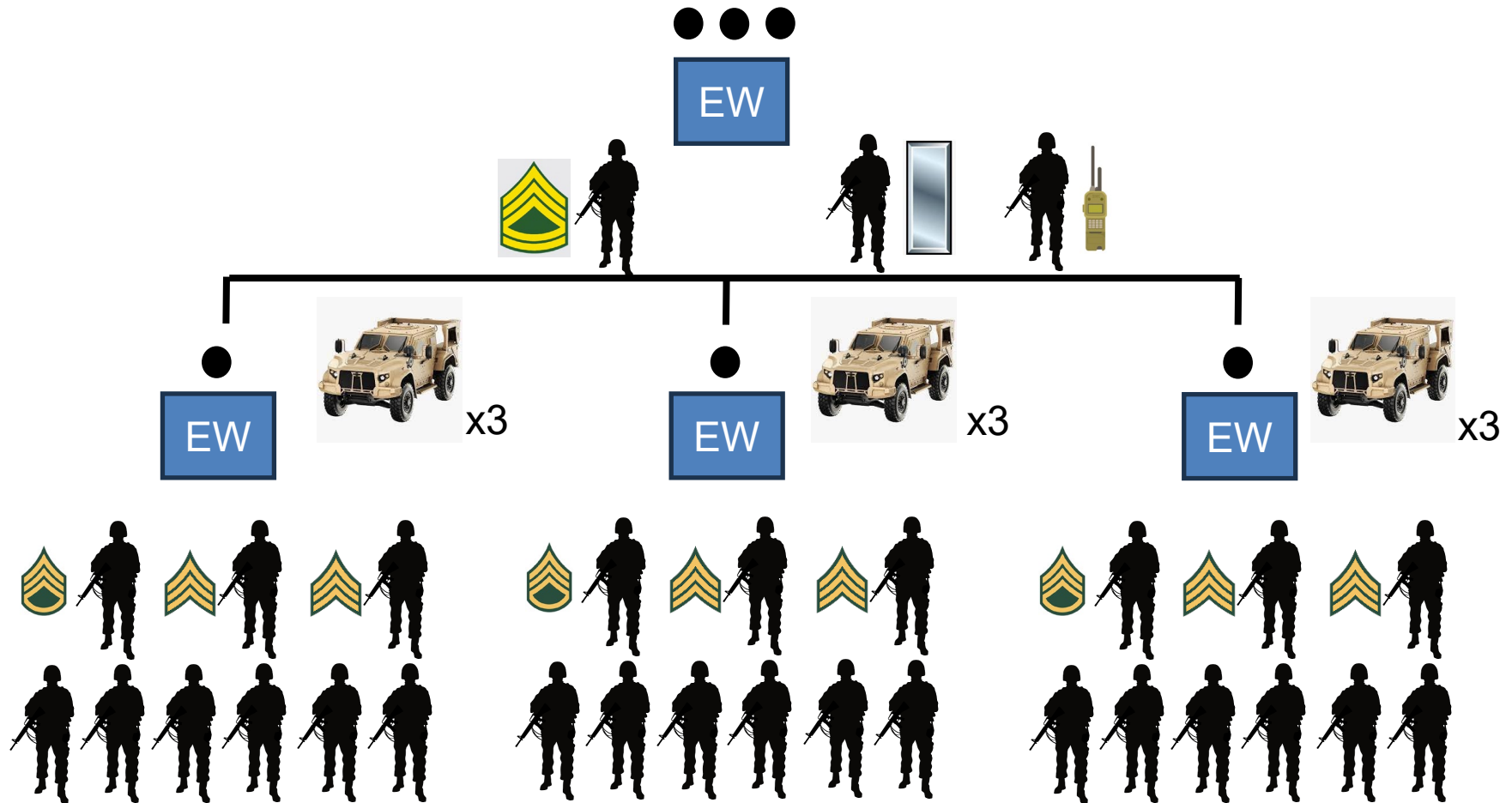




MBCT EW PLT Force Structure



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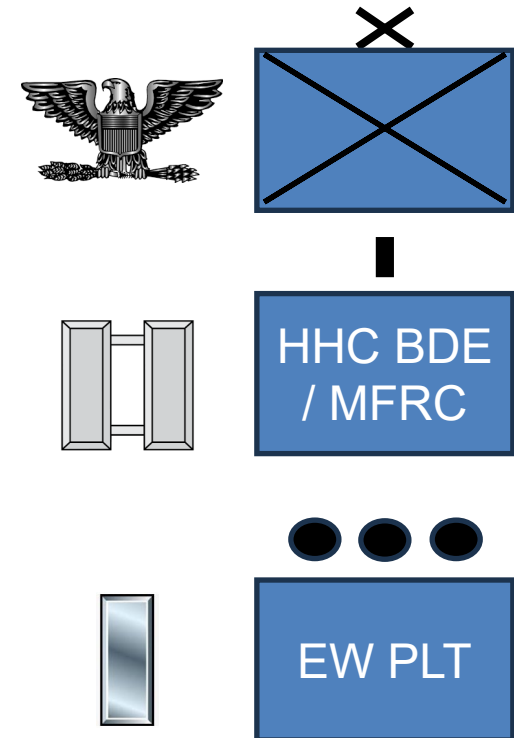


Location within the BDE



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- The Chain of the command for the EW PLT is simple to understand
- The PLT resides in the BDE Headquarters and Headquarters Co for ABCT/SBCT, or the Multi-Functional Reconnaissance Company (MFRC) for MBCT
- HHC BDE & the MFRC company falls under the Brigade HQ, reporting directly to the BCT Commander.
- Supporting relationships for the PLT are much more complicated!





Platoon Leader

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- The platoon leader bears responsibility for all that the platoon does or fails to do.
- The platoon leader is responsible for the collective training, administration, planning, tactical employment, personnel management, and logistics of the platoon.
- The platoon leader must know
 - Capabilities and limitations of personnel and equipment
 - How to tactically employ the platoon against threat organizations, doctrine, and equipment.



Nesting and Supporting Relationships



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- Platoon is organic to BDE Headquarters and Headquarters Co or the Multi-Functional Reconnaissance Company (MFRC)
- Under administrative and garrison control of the Brigade Headquarters
- Under the operational control of the Brigade S-3 or a designated OPCON unit
- Under the technical control of the Brigade CEMA cell
- Supports and coordinates with the Brigade S-2
- Supports and coordinates with the Brigade Fires or the OPCON BN fires section



Platoon Sergeant

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- The platoon sergeant is second-in-charge and the most senior enlisted member in the platoon.
- During operations, the platoon sergeant leads elements of the platoon as directed by the platoon leader.
- The PLT Sergeant must:
 - Assist and advise the platoon leader.
 - Maintain responsibility development of subordinate NCOs
 - Maintain accountability for PLT training, discipline, and welfare
 - Assist other platoon members, as necessary.



EW Teams



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- EW teams' tasks and missions require team members to integrate their efforts into accomplishing crew tasks that an individual Soldier cannot perform.
- Due to the small size and diverse tasks of the EW team, each team member must be familiar with all team tasks and ready to assume the duties of other team members or the team chief.





Team Chief

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- The EW team chief is responsible for mission accomplishment, personnel/team training, counseling, maintenance, accountability, and proper operation of equipment.
- Duties include
 - Training and discipline of the team.
 - Tactical employment and control of the team.
 - Maintenance and operation of assigned vehicles and equipment.
 - Monitoring and reporting status of personnel, weapons, and equipment.
 - Distribution of ammunition, equipment, and loads.
 - Assisting in development of the scheme of maneuver.



Team Chief

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- Planning and establishing hasty or deliberate listening posts.
- Calling for and adjusting indirect fires as required.





Team Members



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- Team members must maintain proficiency and expand personal competence in their individual military occupational specialty tasks through individual, self-directed, and collective training events.
- Tasks include
 - Practicing vehicle self-recovery and combat lifesaver tasks.
 - Performing field maintenance on assigned EW systems.
 - Establishing fighting positions.
 - Filling and operating assigned radios and EW equipment.
 - Establishing hasty or deliberate listening posts.
 - Maintaining situational awareness of EMS emissions.



Team Members



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- Detecting, classifying, and reporting threat emissions and signals of interest.





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Define EW PLT Mission



EW PLT Mission

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- EW platoons conduct electromagnetic support and electromagnetic attack in close contact with enemy organizations and civilian populations to allow maneuver commanders to seize, retain, and exploit the initiative and obtain positions of relative advantage.





Electromagnetic Support



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- When conducting electromagnetic support, the platoon provides indications and warnings, survey of the electromagnetic spectrum, radio frequency direction finding, and geolocation of threat emitters to enable situational understanding of the electromagnetic spectrum for decision making and targeting.
- Electromagnetic support aids in:
 - Answering the commander's critical information requirements.
 - Refining the high-payoff target list and target support matrix.
 - Supporting information collection, integration, and multisource analysis efforts.

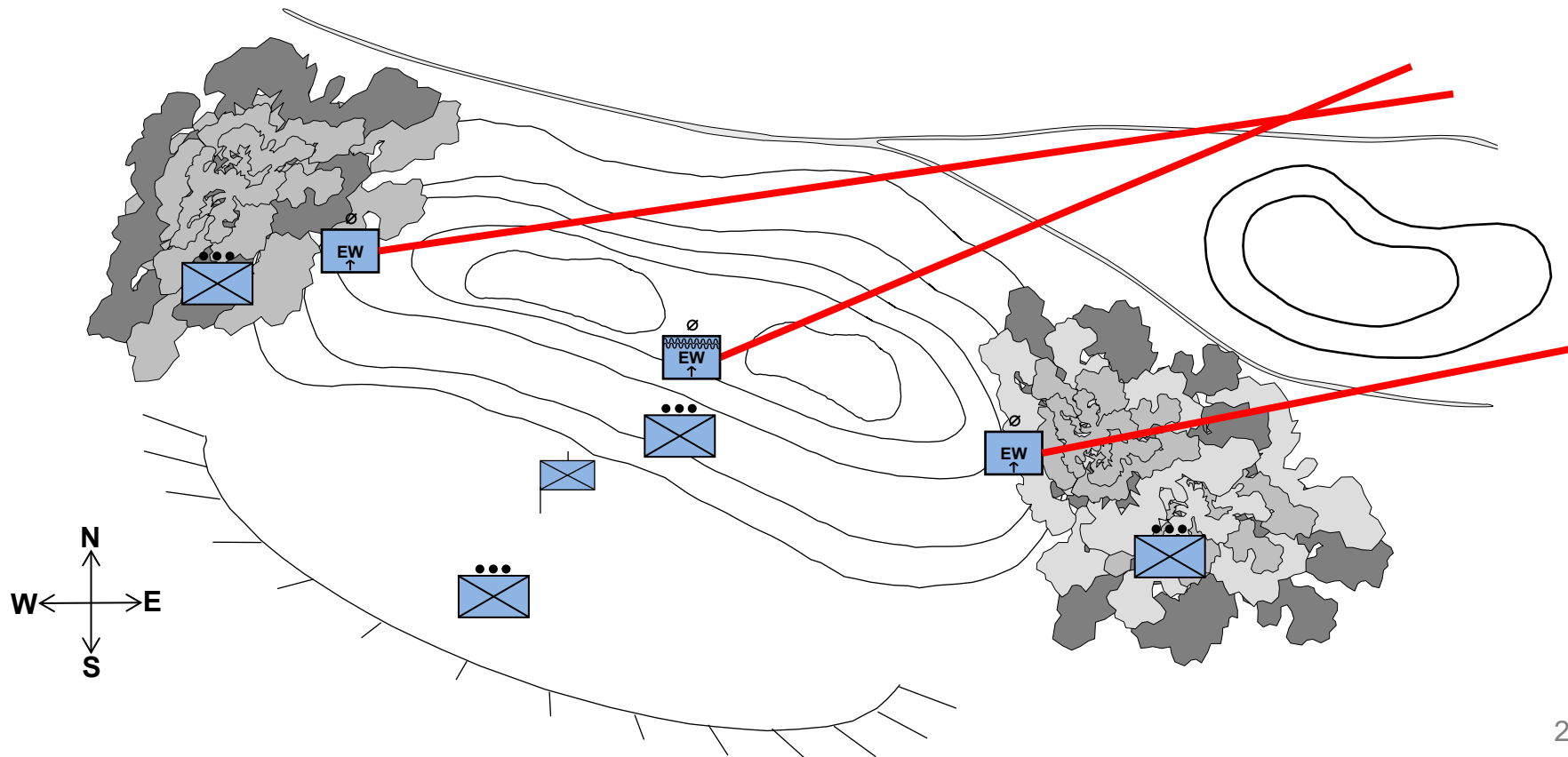


Electromagnetic Support Example



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- EW PLT tasked for Indications and Early Warning during the defense.
- Which avenue of approach will the enemy come from?





Electromagnetic Attack



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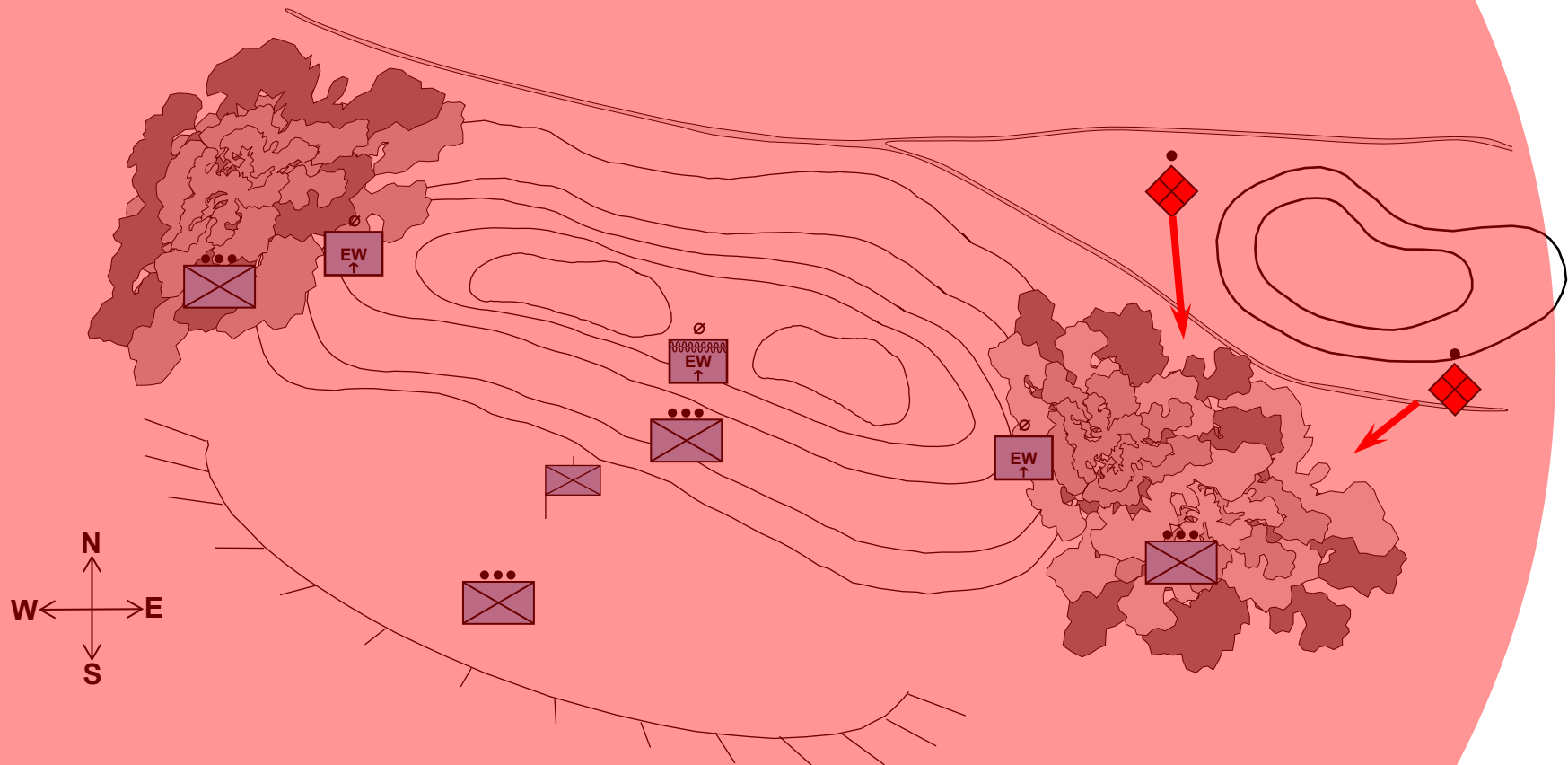
- When conducting an electromagnetic attack, the platoon supports offensive and defensive operations by denying, degrading, disrupting, destroying, or manipulating threat combat capabilities.
- Electromagnetic attack can also support psychological operations, civil affairs, and cyberspace effects.



Electromagnetic Attack Example

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- EW PLT tasked for comms EA upon enemy contact





Electromagnetic Protection



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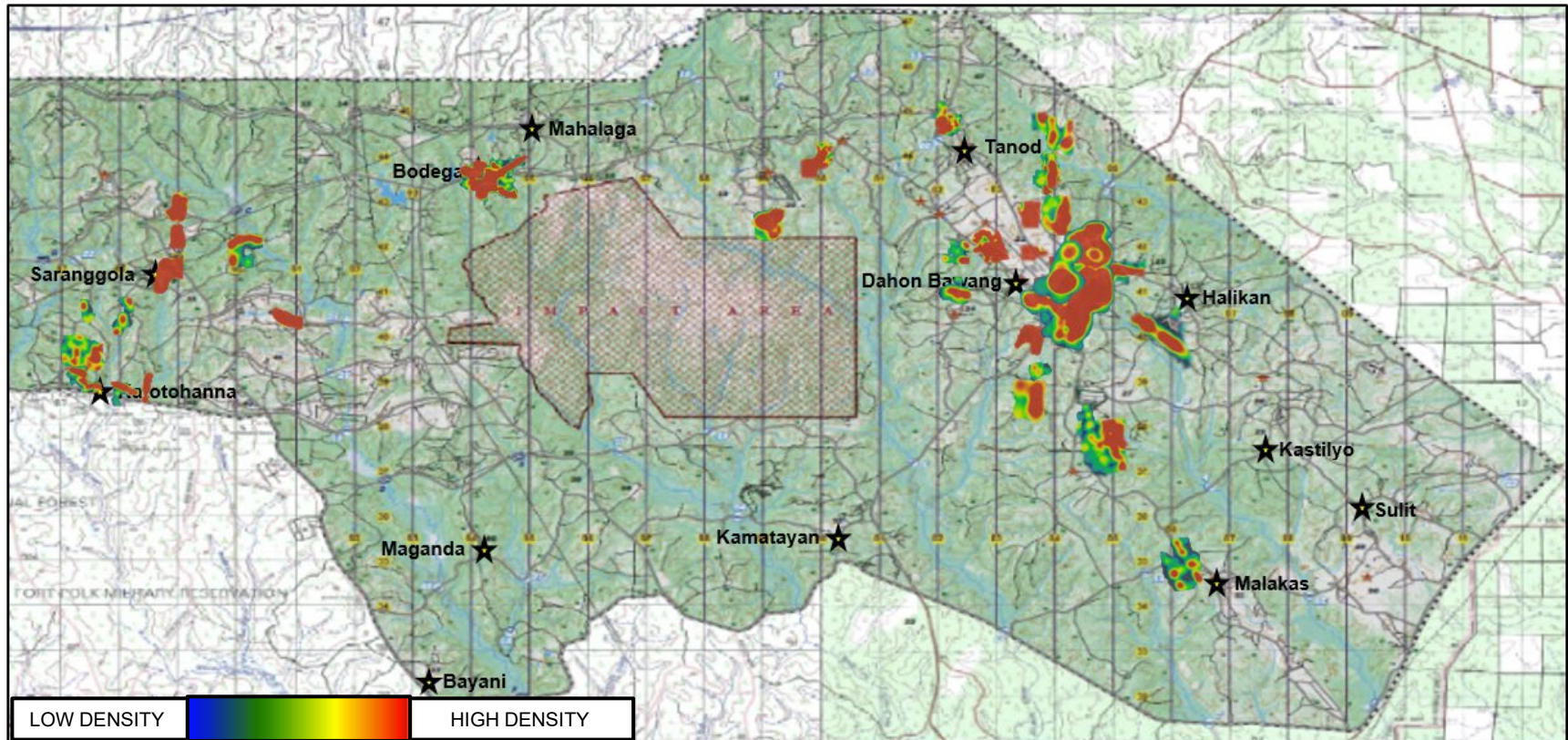
- When supporting electromagnetic protection, the platoon senses and assesses friendly emissions and electromagnetic signatures to enable the commander, G-3 or S-3, and G-6 or S-6 staff to plan appropriate electromagnetic protection and other force protection measures against unintentional and intentional electromagnetic interference, geolocation, and indirect fires.



Electromagnetic Protection Example



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Total Detections

15,293

Each one of these detections denotes a 10-digit grid to an RTU signature.



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Electromagnetic Protection Example



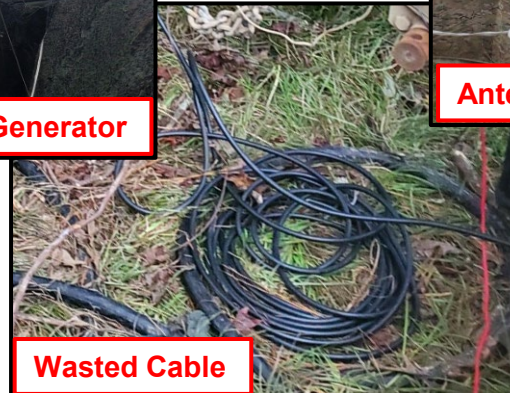
11 Antennas within 20m



Antennas by Generator



Antennas within Sleep Area



Wasted Cable



Key Associated Tasks

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- The EW platoon is manned, trained, and equipped specifically to provide capabilities for its supported CDR.
- Capabilities include:
 - Establishing EW sensing and electromagnetic attack sites.
 - Conducting radio frequency surveys of the environment.
 - Direction finding and geolocating radio frequency emitters.
 - Providing indications and warnings.
 - Supplementing SIGINT collection, as required.
 - Conducting EA against the enemy.
 - Developing the signals of interest and frequencies of interest lists.
 - Enabling cyberspace operations.



Key Associated Tasks



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- Supporting manipulation and electromagnetic deception.
- Supporting development of the electromagnetic order of battle as needed.





Vulnerabilities/Limitations



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- By itself, the EW platoon is vulnerable during operations.
- EW PLTs may require augmentation from the supported unit to adequately defend remote sites against a largescale attack.
- The platoon and its teams should avoid direct engagement with enemy forces and be prepared to displace to an alternate site if the threat situation changes.
- Vulnerability to direct and indirect fire.
- Limited dismounted capability.
- May require augmentation or reach back support for analysis/coordination of ops.



Vulnerabilities/Limitations



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- May have limited authorities to conduct certain EW activities during noncombat operations.
- May have limited authorities or may need to seek additional authorities to conduct SIGINT operations or support to SIGINT operations.



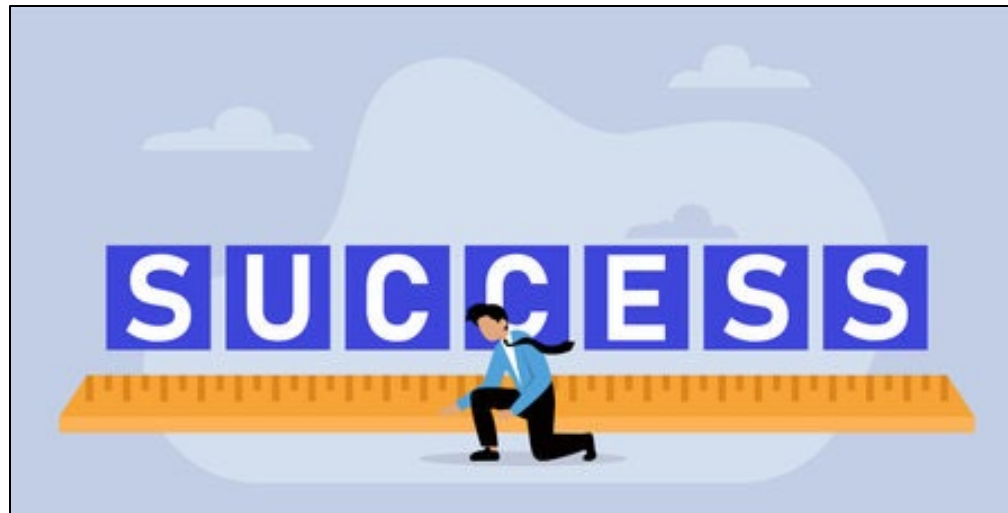


The "So What" of EW PLTs



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- EW PLTs support combat operations through ES, EA and EP by providing targetable/actionable information as well as measurable and achievable effects in the EMS.





The "So What" of EW PLTs



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• What It Can Do



- Sense Signals
- Provide Geo Location to Signals
- Conduct EMS Base Line
- Conduct Localized EA (Jamming)
- Provide SA in the close area

• What It Can't Do



- Break Radio encryptions*
- Sense an entire BDE AO
- Jam an entire AO
- Sense at an infinite distance



The "So What" of EW PLTs



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- What It Should Do 
- Be placed near or beyond the FLOT
- Operate with other reconnaissance elements
- Provide actionable data for targets of opportunity in a NAI and give warnings and indicators of PIRs.
- Tip and Cue other sensors and systems
- Provide general EMS data for a given location (jamming indicators/RF noise level)



The "So What" of EW PLTs



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- What It Shouldn't Do 
- Be used as a primary collection system
- Be assigned an entire AO for sensing
- Jam outside of emergency or decisive situations
- Be placed at the rear or near CPs
- Target RADARs
- Be a long-term collection assets

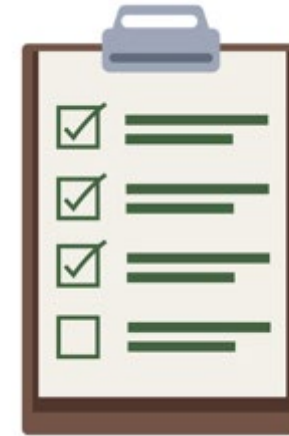


The "So What" of EW PLTs



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- What It Needs to do to succeed
 - Be Semi Autonomous
 - Provide internal local Security
 - Have a reliable and validated PACE Plan
 - BLOS (Data)
 - Move to and Establish sites Tactically
 - Understand, Operate and Maintain EW equipment
 - Understand Targeting and Reconnaissance processes and requirements
 - Not be a burden to supported units.





The "So What" of EW PLTs



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- What It Needs from the Unit
 - Extended Logistical Support
 - Area Security
 - An assigned NAI or Local AO
 - Refined PIRs
 - Target attack guidance matrix
 - EOB with Specific SOIs
 - Clear guidance and CDRs intent
 - Clear and Concise Targeting and reconnaissance requirements





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Compare EW PLT Missions by organization



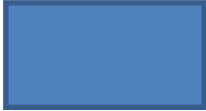
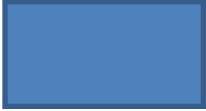
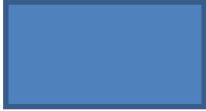
Unit Echelons with EW



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Operational

Tactical

	× × × ~50,000	× × ~15,000	× ~3,000
Echelon	 	 	 

EW Staff

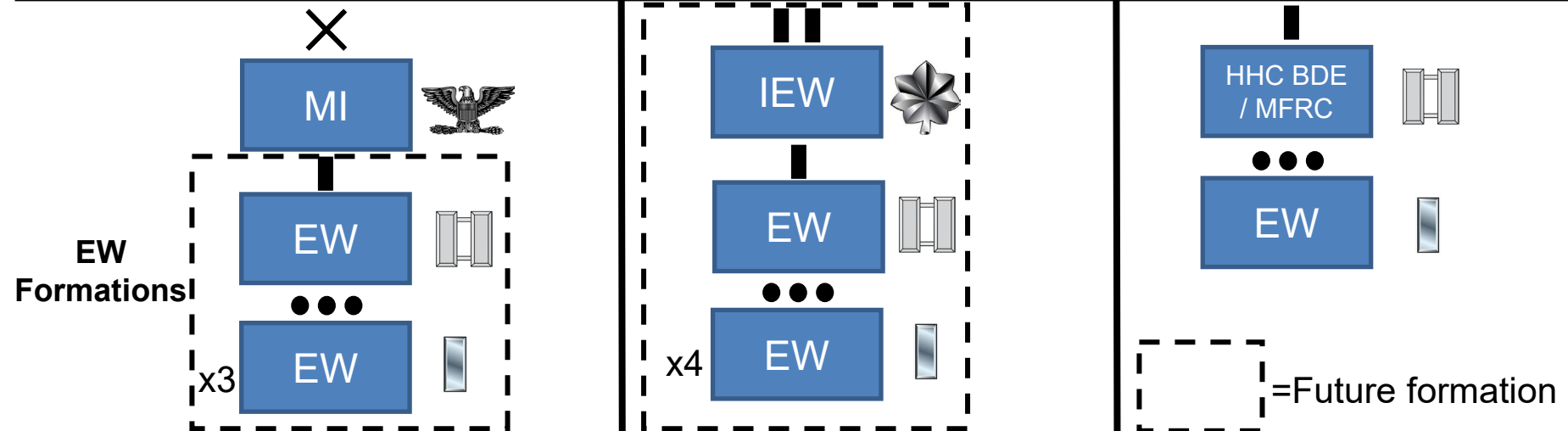
CEMA



CEMA



CEMA



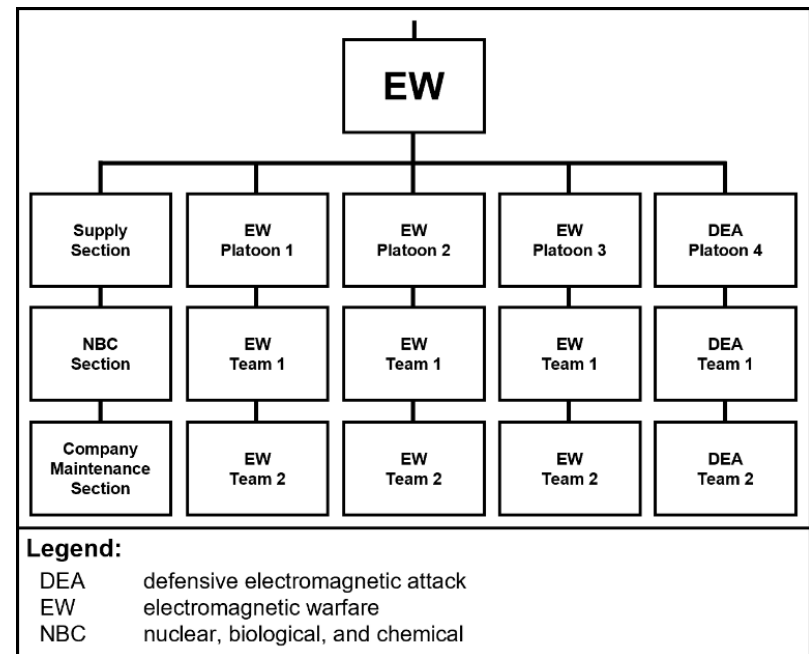


Division EW Company



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- Organic to the division intelligence and electronic warfare battalion.
- Support DIV scheme of maneuver through EA and ES.
- Disperses PLTs throughout the DIV AO.
- Primarily concerned with close area support.



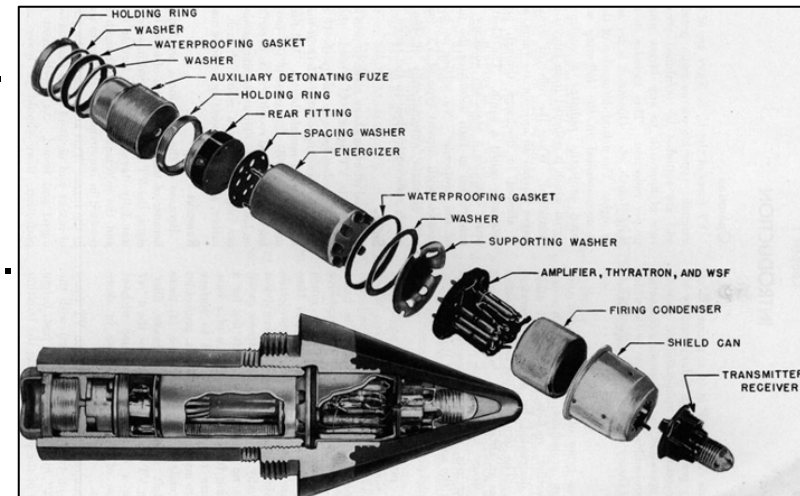


Division EW Company - Defensive EA PLT



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- The platoon provides electromagnetic support and defensive electromagnetic attack to defend against enemy spectrum-dependent equipment, including:
 - Artillery fuses.
 - Radio-controlled improvised explosive devices.
 - Spectrum-homing rockets and missiles.
 - Observation devices.
 - Probing—physical and electromagnetic.
 - Other electro-optical devices.



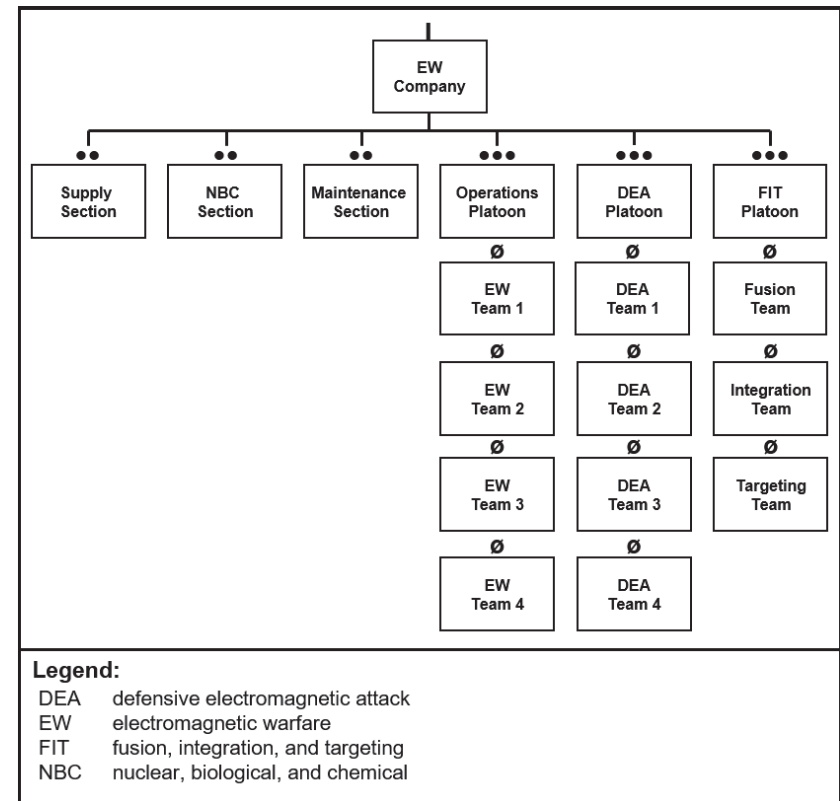


CORPS EW Company



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- Organic to the CORPS MI BDE
- Support CORPS scheme of maneuver through EA and ES.
- Uses Long range sensors and platforms.
- Primarily concerned with deep fight.





CORPs EW Company – Operations PLT



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- The operations platoon provides planning, management, coordination, and oversight of EW activities for all elements.
- The platoon leader advises the commander on capabilities, authorities, and policies associated with EW activities.



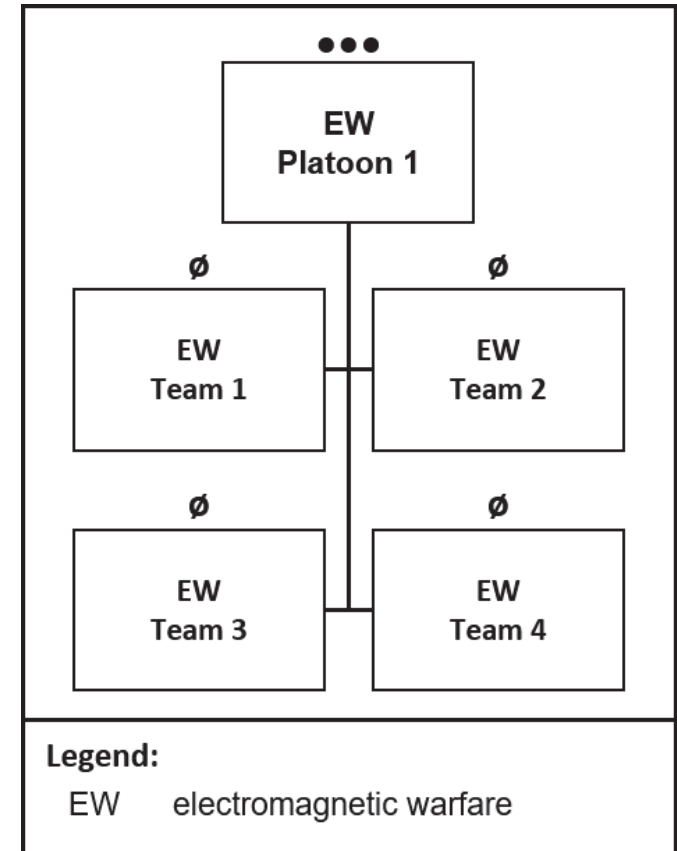


SOF EW PLTs



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- Special forces group EW teams generally deploy independently of the platoon; the platoon leader exercises administrative control of the platoon's EW teams and supports planning activities for deployed elements during distributed operations
- Each special forces group has an organic EW PLT
- PLT provides EA and ES capabilities
- Supports the CDR S-3 and S-6 staff in developing EP measures



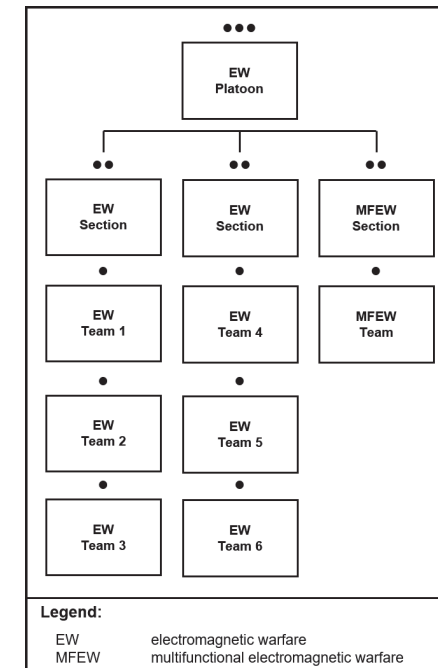


Muti Domain Task Force EW PLT



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- Extended range sensing and effects CO - EW PLT
- Consists of two EW sections and an aerial multi-functional EW section
 - Provides sensing, direction finding, and geolocation.
- Supports:
 - Target development
 - LOS and BLOS EA
 - Aerial LOS reconnaissance for long-range detection
 - Effects within the information dimension that open windows of advantage for the multidomain task force and joint force.





Muti Domain Task Force EW PLT



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- Information Defense CO – Defensive EA PLT
 - The defensive electromagnetic attack platoon provides control, technical support and mission guidance to subordinate EW teams.
 - The platoon headquarters consists of a platoon leader and a platoon sergeant.



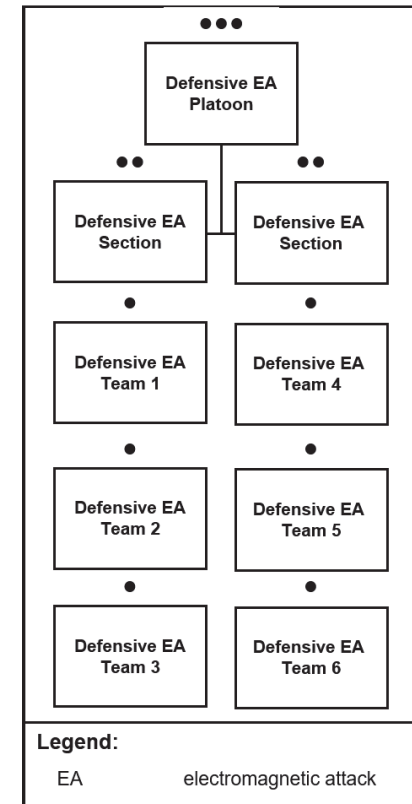


Muti Domain Task Force EW PLT



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- Defensive EA Section
 - Provide ES and Defensive EA to critical nodes across the MDTF's area of operations.
 - Each section consists of three Defensive EA teams.
- Defensive EA Teams
 - Support cruise missile defense, counter unmanned aircraft systems, and counter-fuse operations through EA.





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Identify KEY EW PLT Doctrine



Important Army Publications for CEMA/EW



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- FM 3-12. Cyberspace and Electromagnetic Warfare Operations.
 - Provides tactics and procedures for coordinating, integrating, and synchronizing Army CEMA operations.
- ATP 3-12.3. Electromagnetic Warfare Techniques.
 - Provides doctrinal guidance and direction to the Army for conducting electronic warfare during multidomain operations.
- ATP 3-12.4. Electromagnetic Warfare Platoon.
 - Provides doctrinal guidance for leaders responsible for planning, preparing, executing, and assessing operations of electromagnetic warfare platoons.



ATP 3-12.4

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- ATP 3-12.4 Provides advance insight into:
 - Role of the EW PLT and it's personnel
 - Key tasks of the PLT
 - Force structure and comparison of various EW formations
 - Nesting of the PLT within tactical operations
 - Core competencies of EW personnel





EW Core Competencies

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- Drive understanding
 - EW professionals gather combat information to characterize friendly, neutral, and threat use of the electromagnetic spectrum and understand the integration of threat emitter system arrays at echelon.
 - This information **enables understanding of friendly vulnerabilities and threat capabilities** while allowing commanders to prioritize and deliver effects.





EW Core Competencies

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- Protect friendly personnel and capabilities
 - EW forces implement and enhance measures to protect friendly personnel, facilities, warfighting platforms, capabilities, and equipment from adverse effects in the electromagnetic spectrum.
 - **EW forces employ measures to mask or control friendly emissions from enemy detection and deny adversaries the ability to locate and target friendly formations.**
 - EW forces detect and mitigate enemy attacks in or through the electromagnetic spectrum to maintain the Army's ability to conduct operations and project power across all domains.

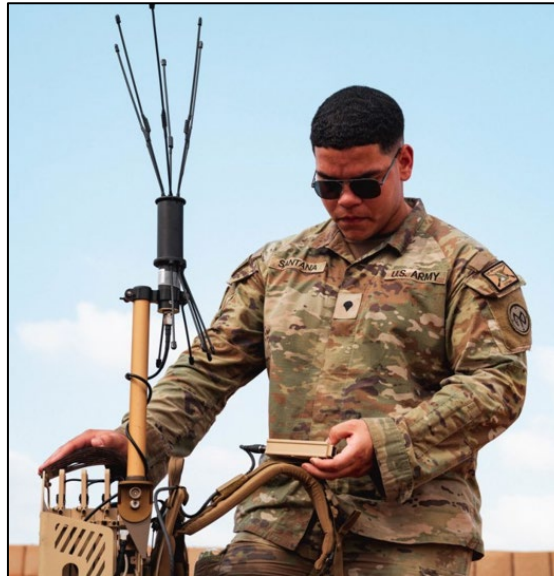


EW Core Competencies



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- Deliver effects
 - EW professionals deliver effects against adversary networks, systems, and weapons through the electromagnetic spectrum.
 - These effects reduce adversary combat power, protect friendly forces, and enhance the lethality of friendly forces and weapons.





EW Fundamental Principles



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- Operational focus
 - EW forces execute missions in support of the supported commander's overarching operational design.
 - EW forces and staff elements integrate and synchronize EW through the operations process as part of a combined arms approach to produce layered dilemmas for the threat in multiple domains and enhance friendly forces' relative combat power.
 - EW staff elements must collaborate across all warfighting functions to be effective.



EW Fundamental Principles



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- Adaptability and versatility
 - EW forces conduct operations using capabilities that are adaptable to various mission requirements.
 - EW capabilities vary in the size of the force employed and the magnitude or scope of effects created.
 - Depending on mission requirements, EW capabilities may be main or supporting efforts for decisive, shaping, or sustaining operations.





EW Fundamental Principles



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- Global reach
 - The nature of the electromagnetic spectrum increases the operational reach of EW forces.
 - EW forces deliver strategic, operational, or tactical effects worldwide from collocated or forward operating positions.



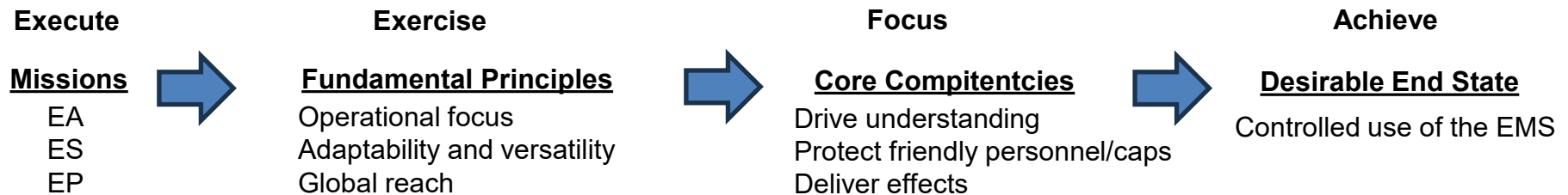


Nesting of Doctrine



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- EW professionals execute missions within the EMS that exercise fundamental principles and maintain focus on core competencies to achieve the desired end state.





Summary



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- Described the EW PLT force structure
- Defined EW PLT Mission
- Compared EW PLT Missions by organization
- Identified Key EW PLT Doctrine



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Questions